

Fundamentals of Solid State Physics

Optical Emission

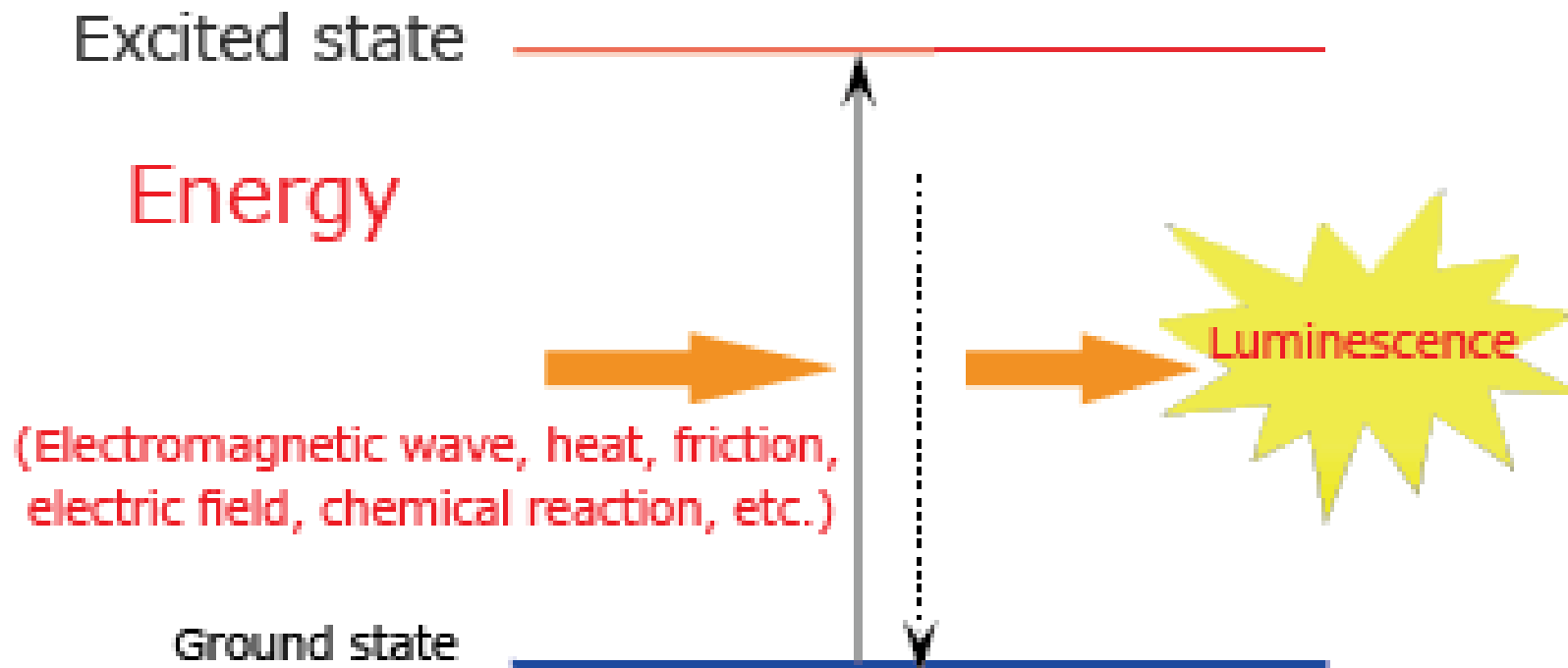
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Optical Emission



Optical Emission

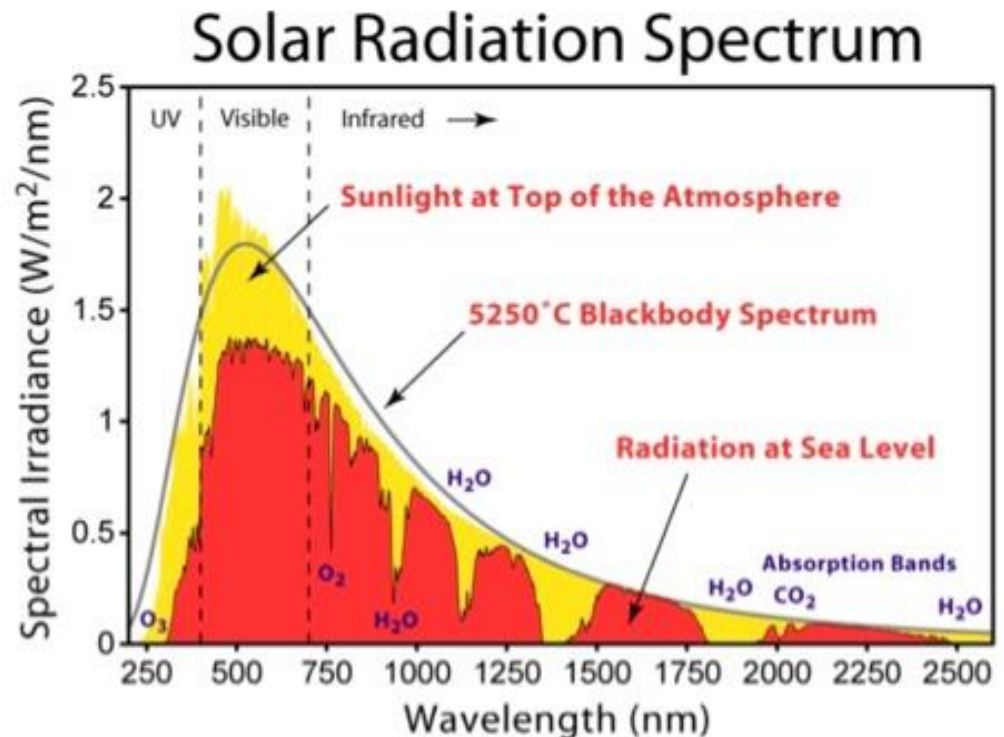
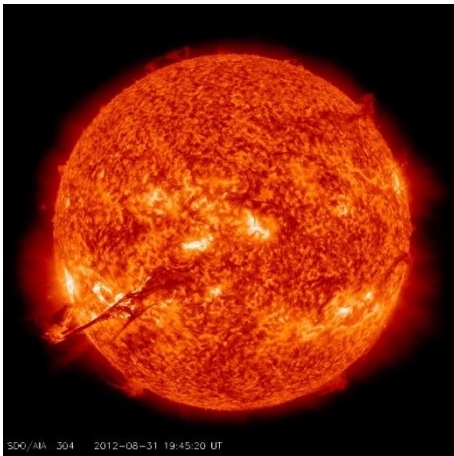
- Thermal radiation 热辐射
- Photoluminescence 光致发光
- Electroluminescence 电致发光
- Others
 - Chemiluminescence 化学发光
 - Bioluminescence 生物发光
 - Sonoluminescence 声致发光
 - ...

Thermal Radiation 热辐射

■ Blackbody Radiation 黑体辐射

- $S(\lambda)$ - radiation power per unit area per unit wavelength ($\text{W}/\text{m}^2/\text{nm}$)

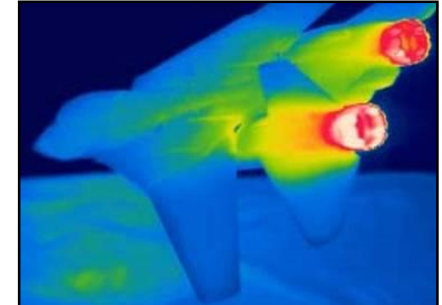
$$S(\lambda) = \frac{2\pi hc^2}{\lambda^5 (e^{hc/\lambda k_B T} - 1)}$$



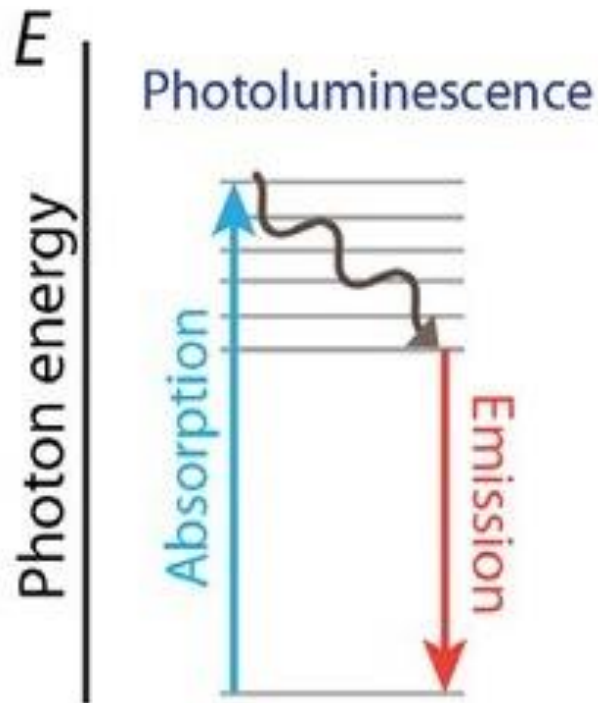
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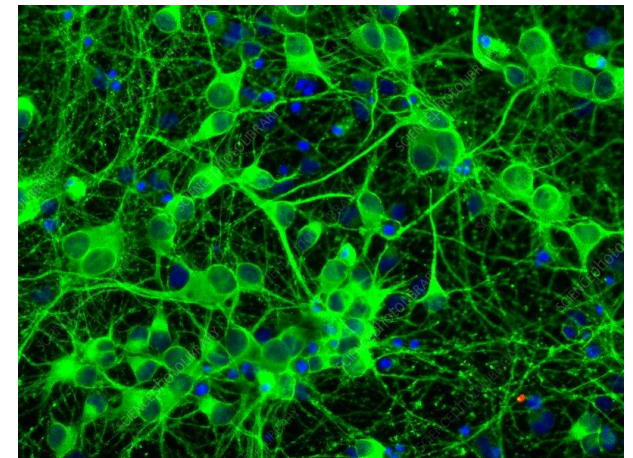
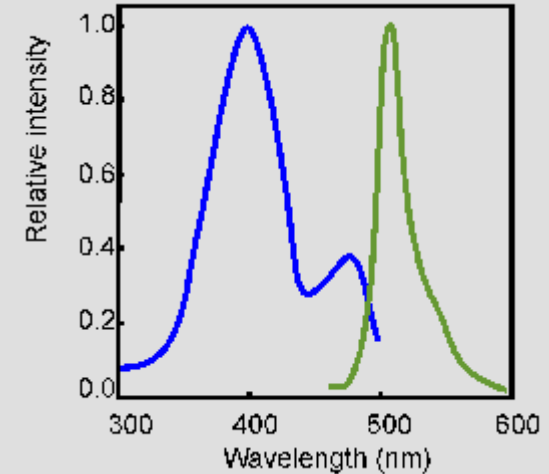
Photoluminescence 光致发光



GFP 绿色荧光蛋白

M. Chalfie, *et al.*,
Science 263, 802
(1994)

absorption emission

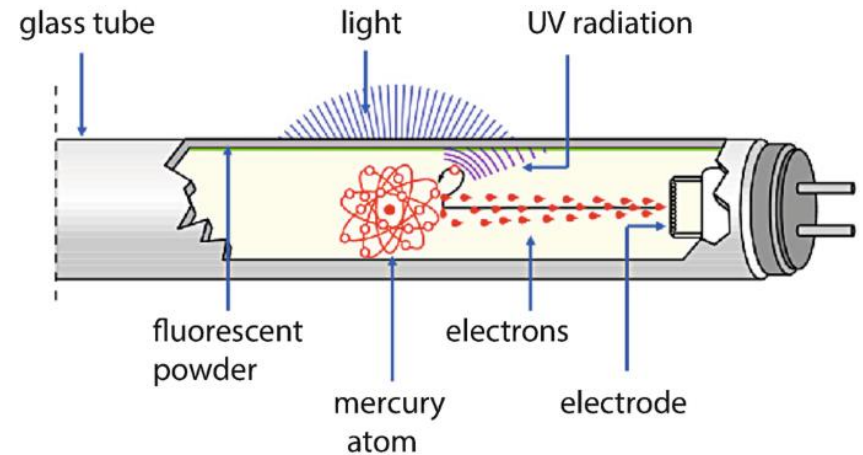


neuron cells

Electroluminescence 电致发光



lightning

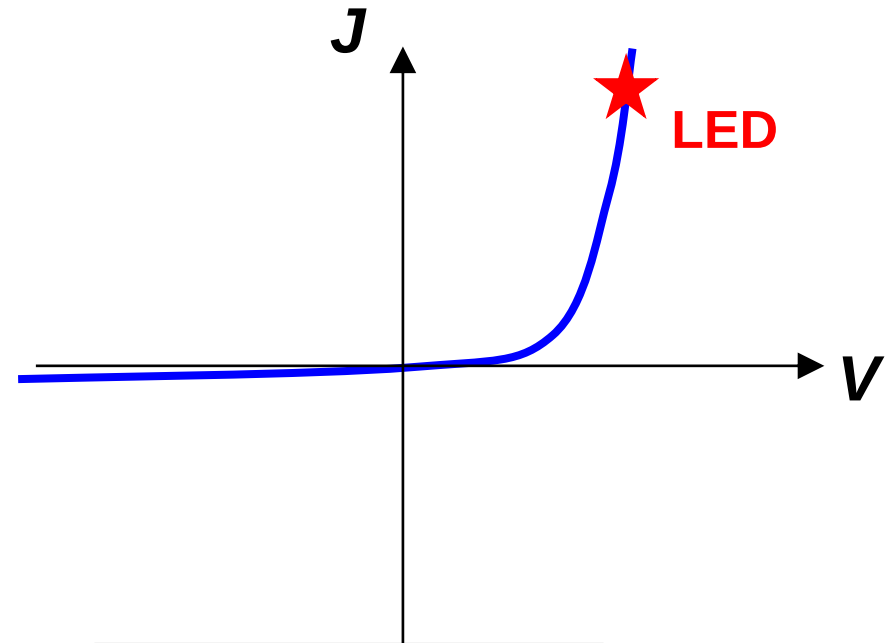
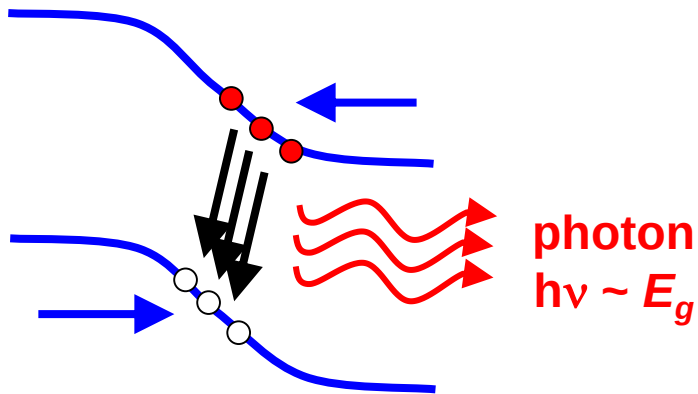


fluorescent lamp



cathode ray tube (CRT)

Electroluminescence 电致发光

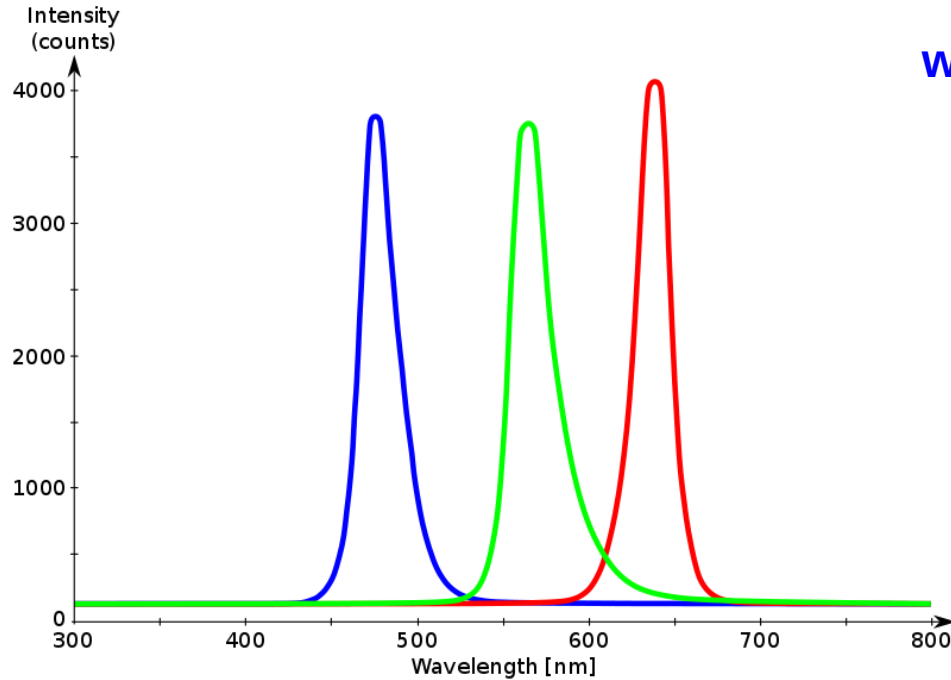
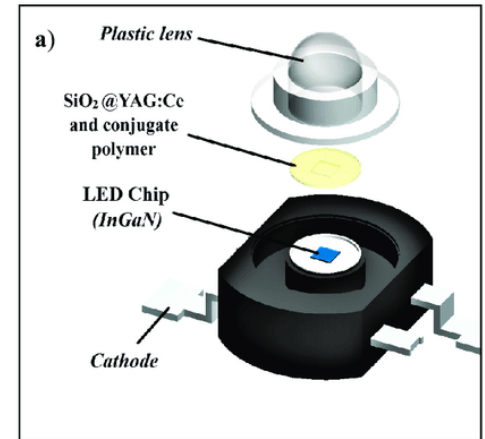


LEDs

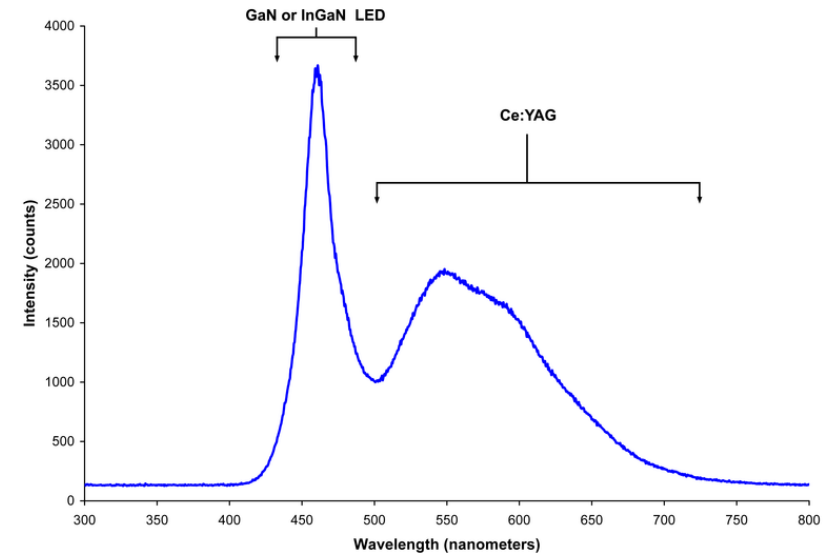


lasers

LED Spectrum

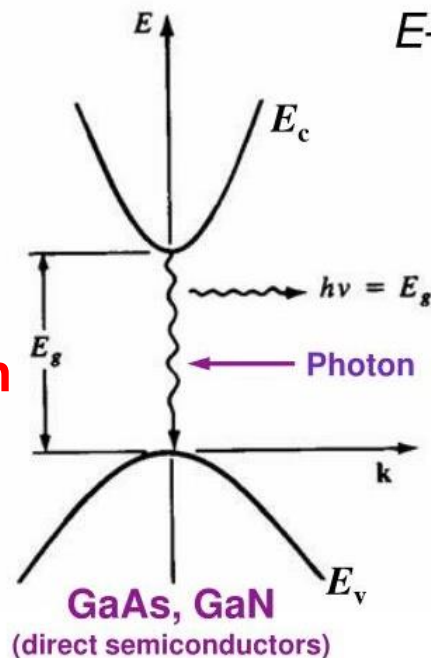


white LED = blue LED + yellow phosphor

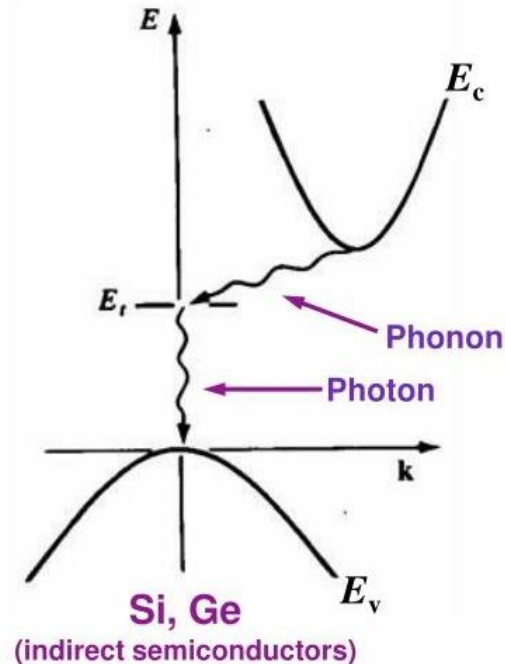


Light Emission Efficiency

radiative
recombination



- Little change in momentum is required for recombination
- Momentum is conserved by photon (light) emission

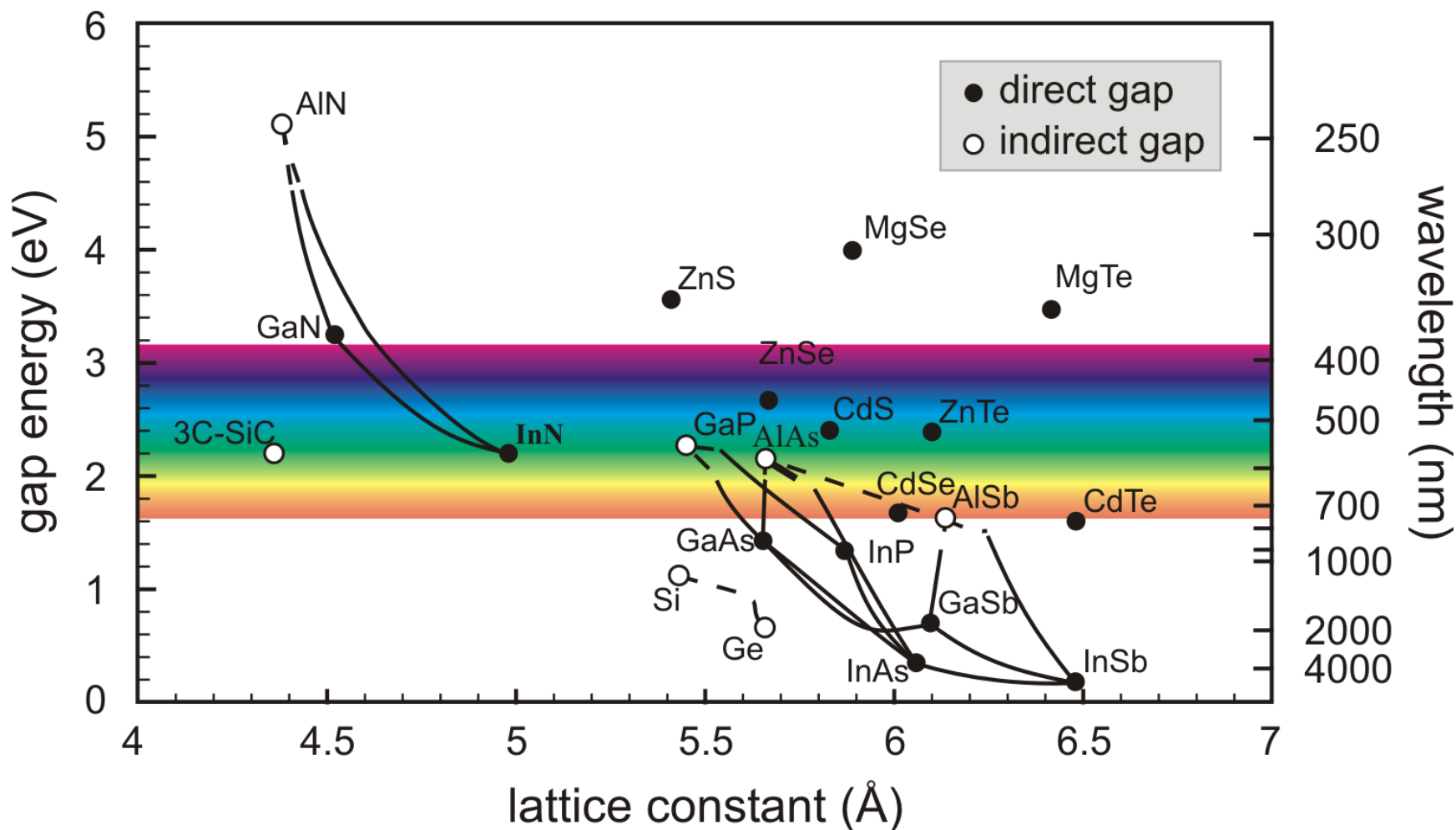


- Large change in momentum is required for recombination
- Momentum is conserved by mainly phonon (vibration) emission + photon emission

Direct bandgap semiconductors like GaAs, GaN are more suitable for LEDs and lasers

Indirect bandgap semiconductors like silicon do not emit light efficiently more non-radiative recombinations

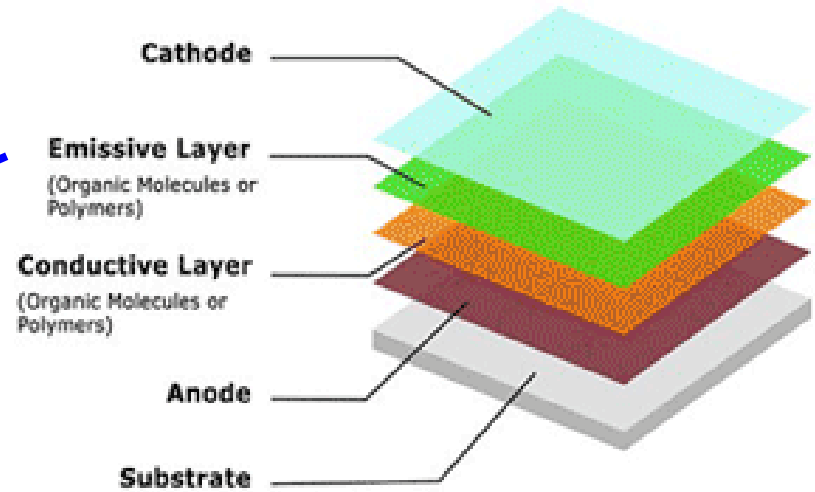
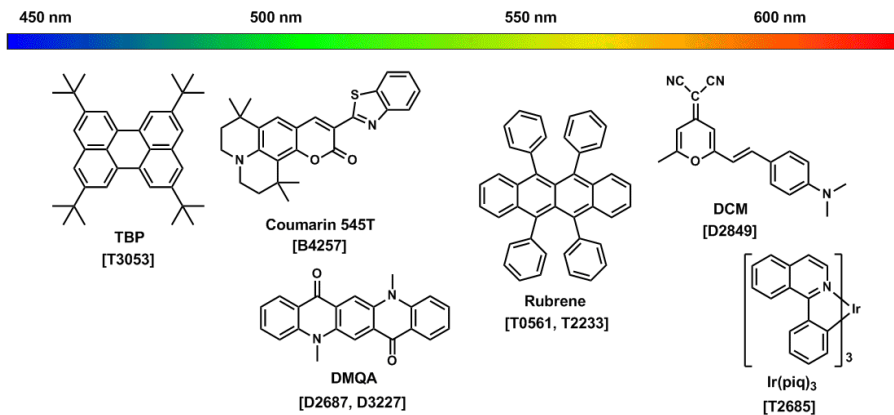
Materials Choices for Light Emission



Organic LED

■ Small Molecules

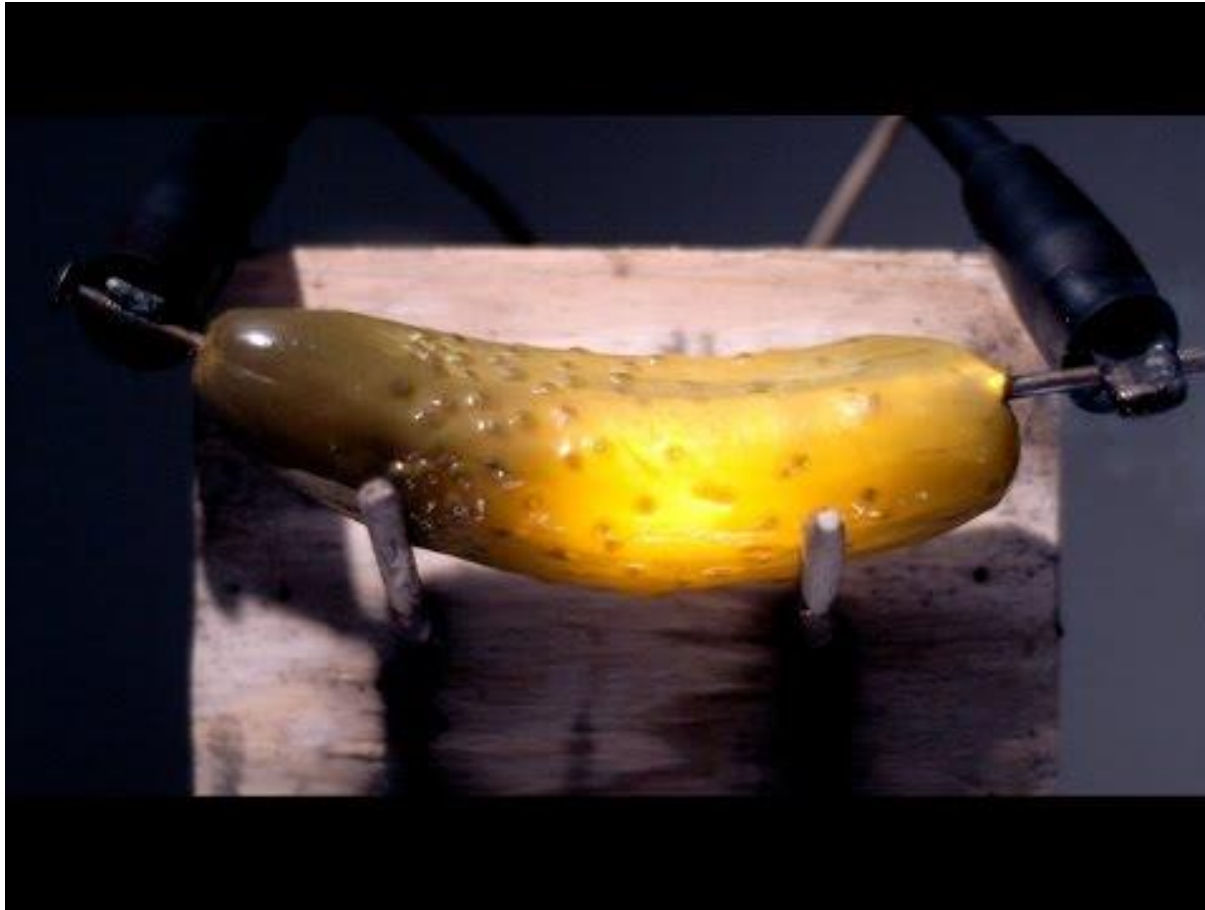
OLED Dopants



OLED



Everything can emit light



Pickle (腌黄瓜)
at 120 V

[Video](#)

Others



Chemiluminescence 化学



Bioluminescence 生物



Others



Chemiluminescence 化学



Bioluminescence 生物



Triboluminescence 摩擦



Sonoluminescence 超声

Lighting 照明技术



Incandescent bulb
白炽灯



Fluorescent lamp
荧光灯



LED lamp

Q: What are the differences?

Display 显示技术



cathode ray tube (CRT)



large screens



laptops



projector



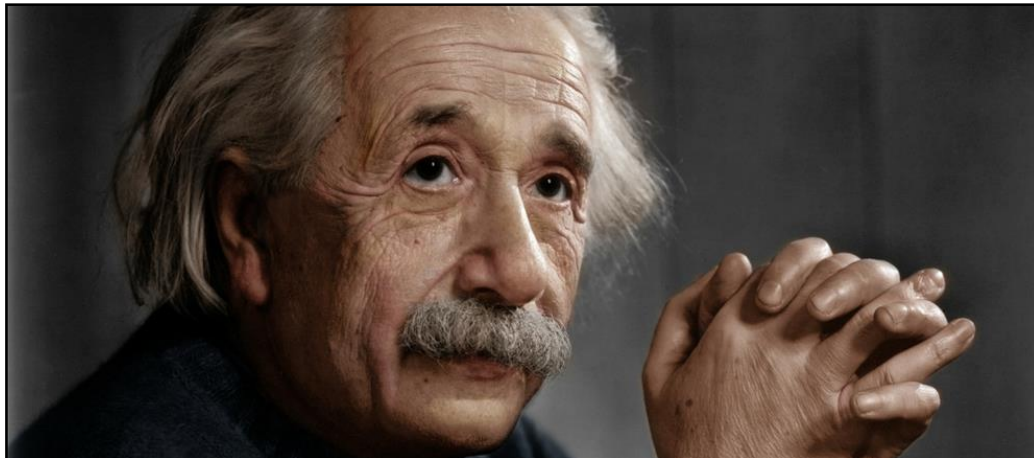
smartphone

Q: What are the differences?

Mysteries Remain

"All these 50 years of pondering have not brought me any closer to answering the question, 'what are light quanta?' "

---- Albert Einstein in 1951



This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- **Optical Properties (Week 14)**
 - ▣ **Origin of Dielectric constant (ϵ) and Refractive index (n)**
 - ▣ **Optical absorption, reflection, refraction, emission**
- Magnetic Properties (Week 15)

Next Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- **Magnetic Properties (Week 15)**
 - **Origin of Magnetics**
 - **Diamagnetism, Paramagnetism, Ferromagnetism**
 - **Superconductivity**

Thank you for your attention